



12 PHYSICS

SPECIAL RELATIVITY

RELATIVITY

Moving Through Spacetime

People are very used to moving through time and space. They are moving through time at the rate of 24 hours every day, but are not moving through space (unless they are in a train).

If a person takes a fast trip to the other side of the world, they have moved through space as well as time.

Einstein's radical idea was that *travel through space and travel through time are interrelated*. In a sense, *the faster a person travels through space, the less they travel through time*. It is as though the person moves through what Einstein called *spacetime* at a constant rate, but the amount of space and the amount of time depend on who measures them.

If someone on Earth observes a space traveller, it turns out that the measurements of her time elapsed and distance travelled do not agree with her measurements. Her time is going *more slowly* than ours. However, the traveller sees that our clocks are going slow. Who is right?

The answer is that *both* are. Time and space are *relative*. But that is what Einstein's theory of relativity is all about.

Frames of reference

Velocity is always measured relative to some particular co-ordinate system or frame of reference.

If you make measurements from a frame of reference that is moving at a *constant velocity*, then any measurements of change in velocity will agree with those made by observers in any other constant-velocity frame of reference.

Frames of reference that move with a constant velocity are called *inertial frames of reference*.

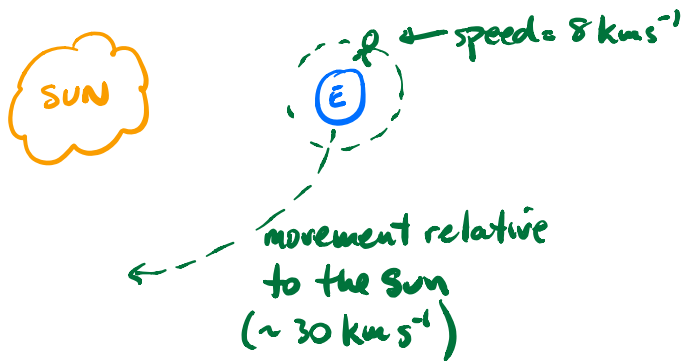
Example

Accelerating reference frames are non-inertial. $\Rightarrow$ General relativity

A car is moving at 60 kmh^{-1} relative to the ground. The assumption is that the car's velocity is measured by a radar gun that is at rest relative to the road.

If the reading is made by a radar gun mounted on a police car following behind at 50 kmh^{-1} relative to the road, the same car is only travelling at 10 kmh^{-1} .

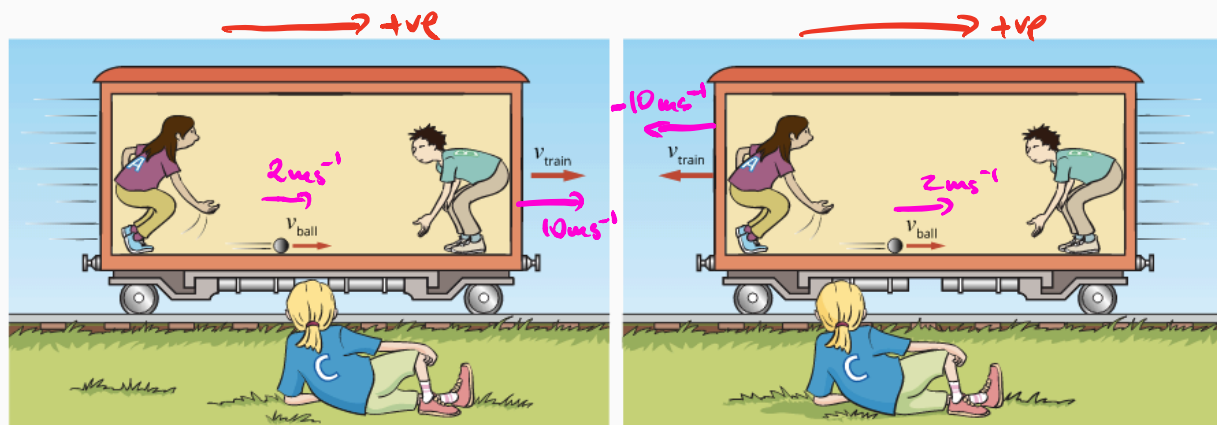
According to Galileo's equations and Newton's laws, *the velocity of objects can be calculated relative to any frame of reference as long as the velocities of the frames of reference are known.*



8.1.4 • This astronaut moves slowly relative to the space shuttle (from where this photograph was taken), at a speed of nearly 8000 m s^{-1} relative to observers on Earth, and at a speed of about $30\,000 \text{ m s}^{-1}$ in the frame of reference of the Sun.

Example

For example, if Amaya and Binh are on a train, and they roll a ball towards the front of the train at 2 m s^{-1} , then relative to the train the ball is moving at 2 m s^{-1} forwards. If Clare is standing still on a platform and sees Binh rolling the ball, then the velocity of the ball relative to Clare depends on the total of the velocity of the train plus the velocity of the ball. If the train is moving at 10 m s^{-1} forwards, then Clare will see the ball moving at 12 m s^{-1} forwards ($10 + 2 = 12$). If the train is moving backwards at 10 m s^{-1} , then Clare will see the ball move at 8 m s^{-1} backwards ($-10 + 2 = -8$), and if the train is stationary, then Clare will see the ball moving at 2 m s^{-1} forwards, just as Amaya and Binh do ($0 + 2 = 2$).



$v_{\text{ball}} = 12 \text{ m s}^{-1} \rightarrow$

$v_{\text{ball}} = -8 \text{ m s}^{-1} \leftarrow$

8.1.5 • Relative to a stationary observer in a frame of reference, the overall velocity of an object moving in a moving frame of reference will depend on the velocity of the object within the moving frame and the velocity of the moving frame with respect to the observer's frame.

SPEED OF LIGHT

The speed of light is one of the most accurately measured constants in physics and its value has been determined very precisely by many different methods.

The accepted value of the speed of light in a vacuum is now:

$$c = 2.99792458 \times 10^8 \text{ ms}^{-1}$$

Physics file: The speed of light



The value of $c = 2.997\,924\,58 \times 10^8 \text{ m s}^{-1}$ for the speed of light is the accepted value, and is now used to define other standards. The speed of light is now the **primary standard** with the unit of length, the metre, being defined as the distance light travels in a time equal to $\frac{1}{2.997\,924\,58 \times 10^8}$ seconds.

This value is consistent with the old standard –the length between two marks on a special bar kept at Sèvres, near Paris. In fact, the distance between the marks cannot be measured to this degree of accuracy.

James Maxwell, working in 1864, derived the speed of light from a mathematical analysis of the propagation of electromagnetic waves.

He took the concept of electric and magnetic fields and worked with them to produce a mathematical description that encompassed all known electromagnetic phenomena. This description can be summed up in four famous equations, referred to as Maxwell's equations.

$$\oint \vec{E} \cdot d\vec{A} = q/\epsilon_0$$

$$\oint \vec{B} \cdot d\vec{A} = 0$$

$$\oint \vec{E} \cdot d\vec{s} = -\frac{d\Phi_B}{dt}$$

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt} + \mu_0 i$$

8.1.9 • Maxwell's equations: The first describes the electric field around a charge in a vacuum. The second describes the fact that magnetic fields are continuous. The third is Faraday's law of electromagnetic induction, which tells you that a magnetic field is generated by a changing magnetic flux. The fourth tells you that a magnetic field is generated by electric currents and/or changing electric flux.

He found that his equations could be used to predict that changing perpendicular electric and magnetic fields could self-propagate through space at a speed given by the fundamental electric and magnetic force constants.

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}$$

Interesting!

where ϵ_0 = the electric permittivity of a vacuum.
 μ_0 = the magnetic permeability of a vacuum.

This speed turned out to be $3.00 \times 10^8 \text{ ms}^{-1}$

Maxwell faced a conundrum, however. His derivation of this speed showed that its value depended only on the two constants for the strength of the electric and magnetic force fields near charges or currents. There was no indication that it should be any different if, for example, the source of the light was in motion.

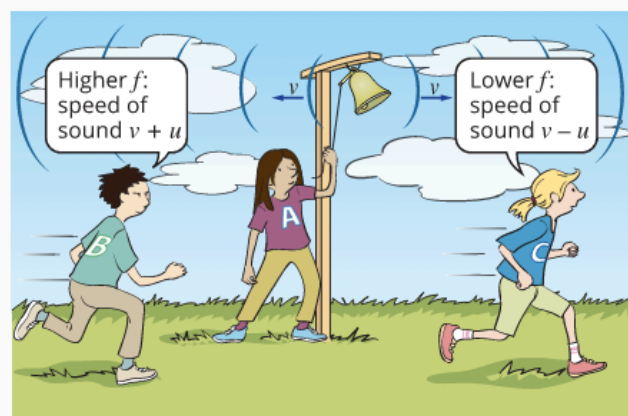
i.e. ***Speed of light is constant in all reference frames.***

There was no indication that it should be any different if the source of the light was in motion.

What was surprising, though, was that there appeared to be no allowance for the speed of the 'receiver' of the light.

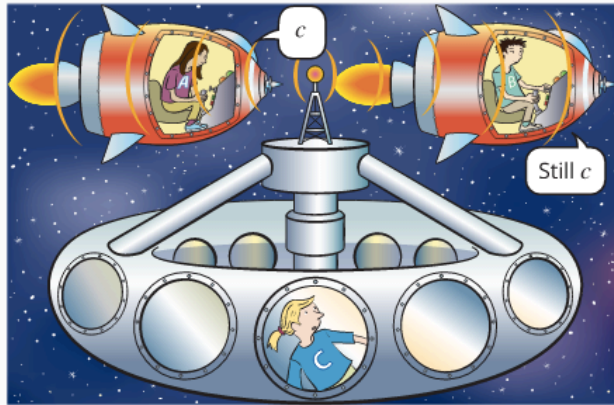
e.g. If you run towards a ball thrown at you, the speed of the ball, relative to you, is greater than if you stand still.

Similarly, if a person is moving towards a source of sound, the speed of the sound (the product of the measured frequency and wavelength) will be the sum of the speed of sound in the medium and the speed of the person.



8.1.10 • As Binh runs towards Amaya, who is ringing the bell, the speed of sound will seem higher than normal. As Clare runs away from Amaya, the speed of sound will seem lower. In both cases, the speed is equal to $f\lambda$, where the wavelength remains the same. However, in the first scenario the frequency will seem higher, and in the second it will seem lower.

Maxwell's equations refused to make any allowance for the motion of the observer of light waves. Whatever the speed of the observer, the measured speed of light is $3.00 \times 10^8 \text{ ms}^{-1}$ - something quite against all the known laws of physics.

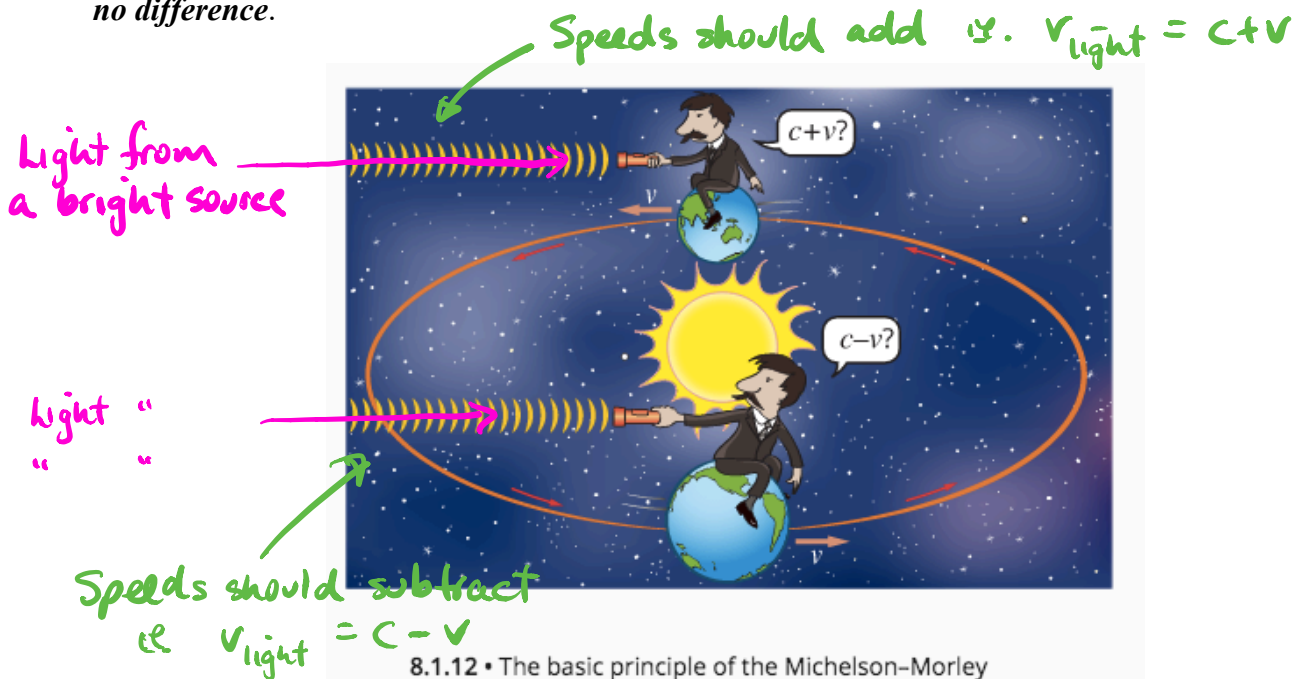


8.1.11 • As Amaya speeds towards Clare's space station and its beacon light, she measures the apparent speed of light to be c . Maxwell's equations state that Binh, as he moves away from the space station, will also find that the speed of light is exactly c .

The Michelson-Morley Experiment

Appreciate the problem.

In the 1880's, Albert Michelson and Edward Morley set up a device known as an interferometer, which was able to measure the very small differences in the time taken for light to travel in two mutually perpendicular directions. They were able to rotate the whole apparatus and hoped to detect the small difference that should result from the fact that one of the directions was to be the same as that in which the Earth was travelling and the other at right angles. However, **they found no difference.**



8.1.12 • The basic principle of the Michelson-Morley experiment. If the aether is fixed relative to the Sun, and the light is travelling (at c relative to the aether) in the same direction as the Earth, the apparent speed should be less than c . Six months later, light travelling in the same direction should appear faster than c .

Six months later, when the Earth would have to be travelling in the opposite direction relative to the light coming from the Sun, there was still no difference in the measured speeds.

The Problem of the Aether

Scientists believed that light travelled through space in a medium - they called it the **aether**. This is a massless and rigid medium that carries both electric and magnetic fields (light).

Most Physicists believed Maxwell's value for the speed of light was relative to this medium, and that if the Earth was moving through this medium, then the speed of light measured would have to be adjusted for the speed of the Earth. The Michelson-Morley experiment showed that there can't be a medium in space - the value for c was constant.

Einstein swept away the idea of an aether existing, accepting that electromagnetic waves can travel in empty space without a medium. There was, however, still the conflict between the absolute speed of light and the principle of relativity in moving reference frames.

EXTENSION

The Michelson–Morley experiment

The existence of an aether appeared to be a serious blow for the principle of relativity. It seemed that there might be, after all, a frame of reference attached to space itself. If this was the case, there was the possibility of an absolute zero velocity.

Scientists needed to test the idea of electromagnetic waves moving through the aether. Since the Earth is in orbit around the Sun, an aether wind should be blowing past the Earth. This suggested to American physicist Albert Michelson that it should be possible to measure the speed at which the Earth was moving through the aether by measuring the small changes in the speed of light as the Earth changed its direction of travel. For example, if the light was travelling in the same direction as the Earth, through the aether, the apparent speed should be slower than usual, i.e. at $c - v$ (Figure 8.1.4). It would be as if the light was travelling against an aether ‘wind’ created by the motion of the Earth through it. If the light was travelling against the Earth’s motion, the apparent speed should be faster as it would be travelling with the ‘wind’ at $c + v$ (Figure 8.1.4). The differences would be tiny, less than 0.01%, but Michelson was confident that he could measure them.

In the 1880s Michelson, and his collaborator Edward Morley, set up a device known as an interferometer. The device cannot measure the speed of light but it can detect changes in the speed of light that might have been due

to the aether wind. In fact, the interferometer was used to attempt to measure the very small differences in the time taken for light to travel in two mutually perpendicular directions. They were able to rotate the whole apparatus and hoped to detect the small difference that should result from the fact that one of the directions was to be the same as that in which the Earth was travelling and the other at right angles. However, they found no difference. Perhaps, then, the Earth at that time was stationary with respect to the aether? Six months later, however, when the Earth would have to be travelling in the opposite direction relative to the aether, there was still no difference in the measured speeds. Other people performed similar experiments, always with the same null result. In whatever direction the Earth was moving it seemed to be at rest in the aether. Or perhaps there was no aether at all.

While Michelson and Morley’s results were consistent with Maxwell’s prediction that the speed of light would always appear to be the same for any observer, the apparent absurdity of such a situation led most physicists to believe that some flaw in the theory behind the experiment, or in its implementation, would soon be discovered. Einstein, however, wondered about the consequences of actually accepting their prediction about the speed of light but at the same time holding on to the relativity principle.

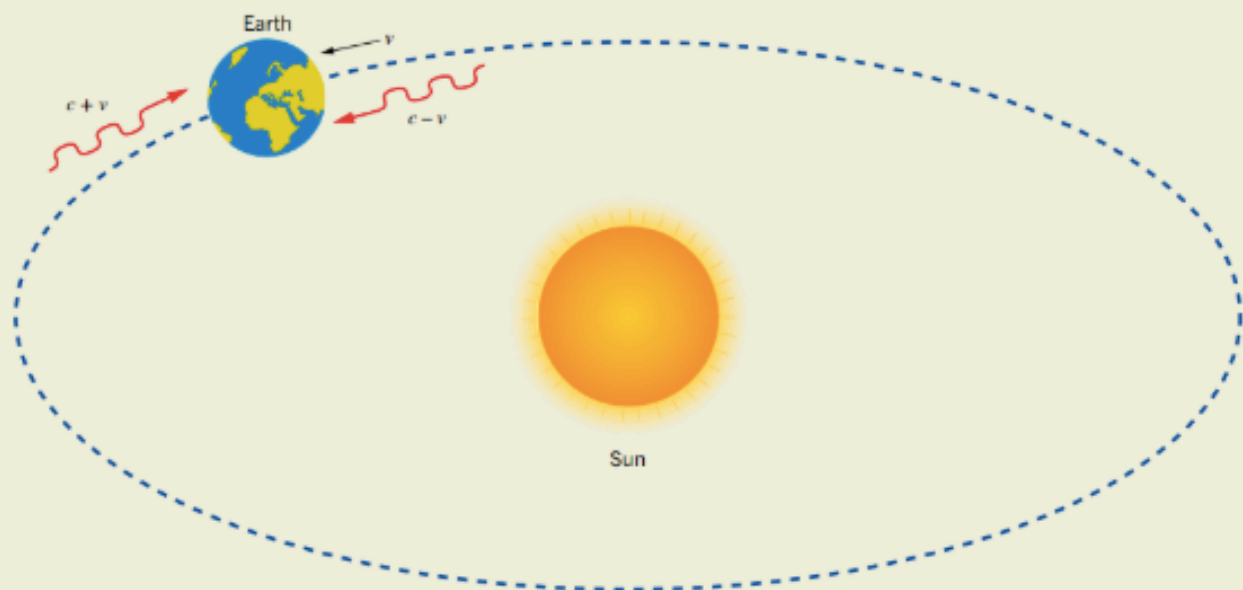


FIGURE 8.1.4 The basic principle of the Michelson–Morley experiment. If the aether is fixed relative to the Sun, and the light is travelling at c relative to the aether and in the same direction as the Earth, the apparent speed should be less than c , i.e. $c - v$. If the light was travelling in the opposite direction to Earth, light should appear faster than c , i.e. $c + v$.

EINSTEIN

Einstein was the archetypal theoretician - the only significant experiments he ever did were *thought experiments*, or 'Gedanken' as they are called in his first language (German). Many of his Gedanken experiments involved thinking of situations that involved *two frames of reference moving with a steady relative velocity*. Newton had referred to these as *inertial frames of reference*, as the law of inertia applied within them, but *not within frames that were accelerating*.

He stated that all inertial frames of reference must be equally valid, and that the laws of physics must apply equally in any frame of reference that is moving at a constant velocity. A consequence of this is that there is no physics experiment you can do, completely within a particular frame of reference, to tell that you are moving.

e.g. As you speed along in a Gedanken train with the blinds down, you cannot measure your speed, at least not without peeking out the window, or connecting something to the wheels, which are in contact with the outside frame of reference.

You can tell if you are accelerating easily enough: just hang a pendulum from the ceiling. However, the pendulum will hang straight down if your velocity is constant, whether you are travelling steadily at 100 kmh^{-1} or are stopped in the station.

THE SPECIAL THEORY OF RELATIVITY

Paradox - how could two observers travelling at different speeds see the same light beam travelling at the same speed?

Einstein put forward two simple postulates.

- The principle of relativity states that no law of physics can identify a state of absolute rest. (In other words, no single inertial frame of reference is better than any other.)
- The speed of light will always be the same no matter what the motion of the light source or observer.

Know these

Newton's laws are based on the idea that space and time are how they seem to a person on Earth - constant, uniform and straight. Space is like a big set of xyz axes that always remain mutually perpendicular, and in which distances can be calculated exactly according to Pythagoras's rule. In this space, **time flows on at a constant rate that is the same everywhere.**

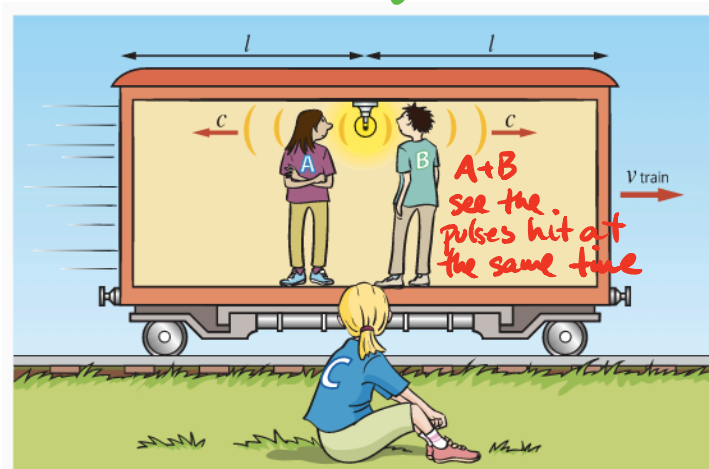
Einstein realised that the assumptions Newton made may not in fact be valid, at least not on scales involving huge distances and speeds approaching that of light. The only way in which his postulates can both be true is if space and time are not 'absolute'.

Einstein's Gedanken Train

Einstein discussed a simple thought experiment to illustrate the consequences of accepting the two postulates. It involves a **train moving at a constant velocity**. **Angela and Bill have boarded** Einstein's train to help us with our experiments and **Clare is still (stationary)** outside on the platform.

This train has a flashing light bulb set right in the centre of the carriage. Angela and Bill are watching the flashes of light as they reach the front wall and back walls of the carriage, respectively. They are not surprised to find that the **flashes reach the front and back walls at the same time.**

This is called simultaneity.

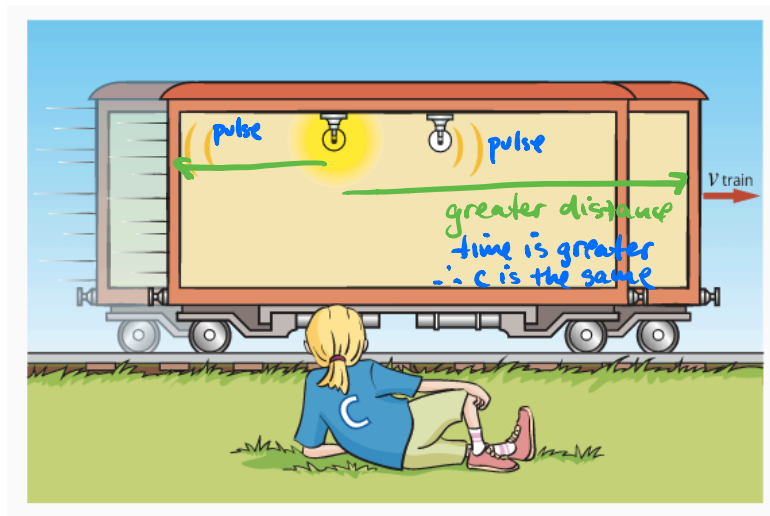


i Einstein's two postulates:

- The laws of physics are the same in all inertial (non-accelerated) frames of reference.
- The speed of light has a constant value for all observers regardless of their motion or the motion of the source.

(The first postulate means that there is no preferred frame of reference and so is sometimes stated as: No law of physics can identify a state of absolute rest.)

Outside, Clare is watching the same flashes of light. Einstein's interest was in when Clare saw the flashes hit the end walls.



The light has to reflect from the walls and then travel to Clare before she can 'see' it hit the wall, but the simplifying assumption made is that she makes the appropriate calculations to determine when the light 'actually' hit the walls - she ignores the **look-back time**.

If Clare sees the light travelling at the same speed in the forward and back direction, **she will see the light hit the back wall first.** This is because that wall is moving towards the light, whereas the front wall is moving away from the light and so the light will take longer to catch up to it.

This is all against the principles of Newtonian physics.

- Angela and Bill saw the light flashes reach the ends of the carriage at the **same time**.
- Clare saw them reach the walls at **different times**.

The idea that two events that are simultaneous for one set of observers are not simultaneous for another is a conundrum.

(Remember - Clare did not see the flashes at different times because of some sort of **look-back time delay** - that has been taken into account.)

This is often referred to as a **lack of simultaneity** - simultaneity means that a person would normally expect that, if one set of observers see two events happen simultaneously, they would expect any others to also see them simultaneously.

Physics file

In discussing relativity, there are often situations where we want to know when light reaches a certain point, like the walls of the train. We (or our intrepid train travellers) only 'see' the light reach a point after it has reflected from that point and returned to our eyes. This extra time is called the **look-back time**. We need to calculate the look-back time and subtract it from the observed time to find the actual time to reach the wall. To avoid confusion, we will assume that our observers always do this and only quote the actual time.

Simultaneity And Spacetime

How is it possible that two events that are simultaneous to one set of observers are not simultaneous to another?

Einstein said that the only reasonable explanation for this is that *time itself is behaving strangely*. The amount of time that has elapsed in one frame of reference is not the same as that which has elapsed in another.

Because time (which has one dimension) seems to depend on the frame of reference in which it is measured, and a frame of reference is just a way of defining three-dimensional space, clearly time and space are somehow interrelated. Physicists call that *four-dimensional relationship - spacetime*.

Special relativity is all about spacetime.

This was a profound shock to the physicists of Einstein's time. Many refused to believe that time was not the constant and unchanging quantity that it had always been assumed to be, and certainly always seemed to be.

And to think that it might 'flow' at a different rate in a moving frame of reference was too mind-boggling for words. That could mean that if a person went for a train trip, their *clocks would go slow* and they should come back *younger* than those who stayed behind.

This is what Einstein postulated, but younger by something much less than a microsecond - rather difficult to notice.

CONSEQUENCES OF SPECIAL RELATIVITY

There are *three consequences* of Einstein's special theory of relativity.

- *Time Dilation* Time measured from outside the moving reference frame is longer.

If Einstein is correct that the speed of light is constant for each observer, regardless of their frame of reference, then one can *compare the period of time for a light event* in two frames of reference.

Example

Angela and Bill are travelling in a Gedanken space ship, while Clare observes from a space station. By observing a photon travelling from the light bulb in the space ship to the floor, one can determine the time taken for the event by dividing the distance travelled by the speed of light.

Angela and Bill see the light travelling the height of the space ship at the speed of light, so:

$$\Delta t_{\text{Angela + Bill}} = \frac{d}{c}$$

At the same time Clare, watching from the space station, sees the same event. The photon travels from the light bulb to the floor at the speed of light, but during that time the *space ship has moved forward some distance* determined by the velocity of the space ship and the period of time for the event. *So Clare sees the photon travel at an angle to the ground.*

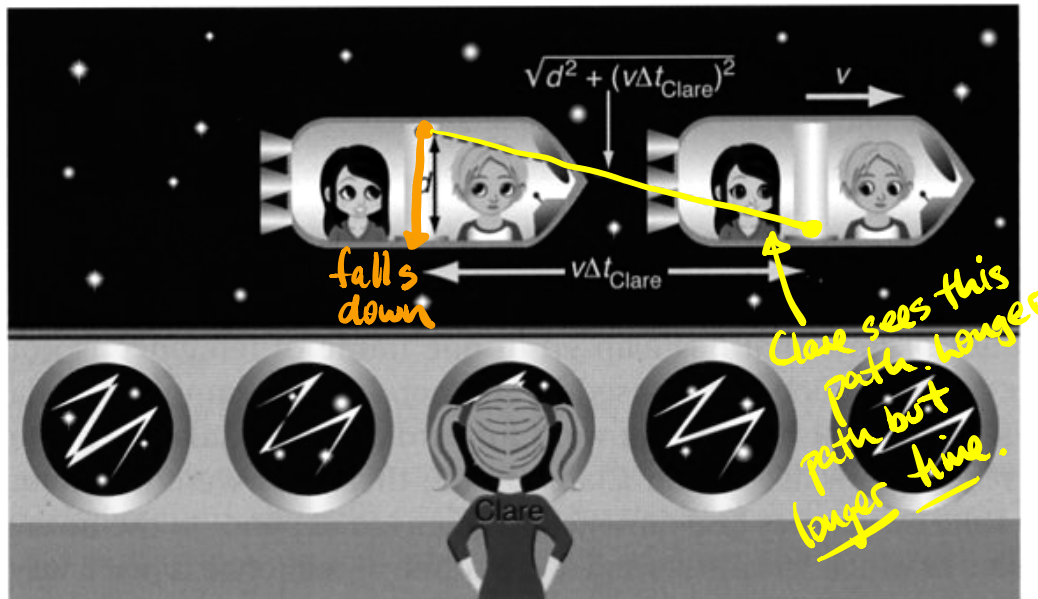


Figure 6.31

A photon event as seen by Clare. The photon has to travel a greater distance at the same speed of light, therefore the event takes a longer time.

The time that the photon takes to travel from the bulb to the floor is:

$$\Delta t_{\text{Clare}} = \frac{\sqrt{d^2 + (v\Delta t_{\text{Clare}})^2}}{c}$$

The extra distance that the photon must travel in Clare's frame of reference means that the *same event takes a longer period of time to occur*.

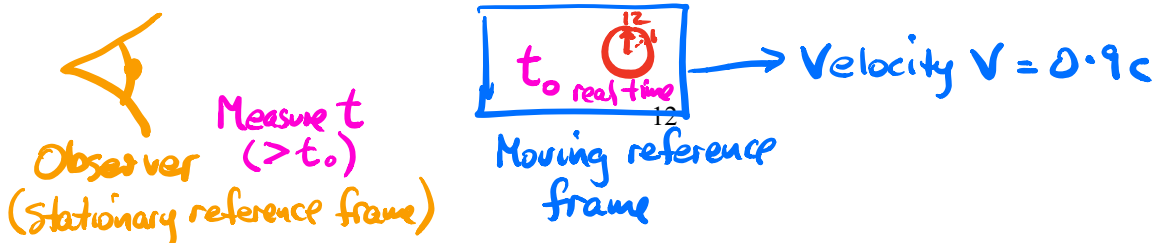
In fact, every event observed in Angela and Bill's frame of reference by Clare will occur over a longer period of time - the time taken for the second hand on Angela's watch to move will take longer, the time it takes for Bill to draw a breath will take longer.

The faster the space ship travels, the longer an event takes to occur, when viewed from a stationary frame of reference.

If Angela and Bill were riding a photon, how long would an event take when travelling at the maximum speed possible - the speed of light? The answer is - forever!

They would appear to Clare to be *frozen in time*, for Angela and Bill everything else would also appear frozen in time, and *it would appear to take no time to travel from one end of the Universe to the other*.

Time dilation is when an event is timed in an inertial (constant non-zero) frame of reference by an observer in another inertial (stationary) frame of reference, and it will occur over a longer period of time than it does for the same event timed by an observer in the moving frame of reference.



To generalise Einstein's equation for time dilation, the symbol t is used to represent the dilated time that a stationary observer measures using a stationary clock, for an event that the observer sees occurring in a moving frame of reference. The symbol Δt is also used to represent this dilated time measured by a stationary observer. The symbol t_0 is then the time that passes on the moving clock, which is also known as the **proper time**. Proper time can also be represented by the symbol $\Delta t'$.

The factor that the proper time is multiplied by is given the symbol γ , so that:

$$t = \gamma t_0$$

$$t = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

where
$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

The physicist H. A. Lorentz first introduced the factor γ in an attempt to explain the results of the Michelson-Morley experiment, so it is often known as the **Lorentz factor**.

Example 1

Assume *Gedanken* conditions exist in this question. A stationary observer on Earth sees a very fast car passing by, travelling at $2.50 \times 10^8 \text{ m s}^{-1}$. In the car is a clock on which the stationary observer sees 3.00 s pass. Calculate how many seconds pass by on the stationary observer's clock during this observation. Use $c = 3.00 \times 10^8 \text{ m s}^{-1}$. 3 parts

Thinking Identify the variables: the time for the stationary observer is t, the proper time for the moving clock is t_0 and the velocities are v and the constant c.	Working $t = ?$ $t_0 = 3.00 \text{ s}$ $v = 2.50 \times 10^8 \text{ m s}^{-1}$ $c = 3.00 \times 10^8 \text{ m s}^{-1}$ <i>Observer Earth</i>
Thinking Use Einstein's time dilation formula and the Lorentz factor.	Working $t = \gamma t_0$ $= \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$
Thinking Substitute the values for t_0, v and c into the equation and calculate the answer t.	Working $t = \frac{(3.00)}{\sqrt{1 - \frac{(2.50 \times 10^8)^2}{(3.00 \times 10^8)^2}}}$ $= \frac{(3.00)}{(0.55277)}$ $= 5.43 \text{ s}$

For normal speeds, the Lorentz factor is essentially 1.000. As the object approaches relativistic speeds, the factor becomes very significant.

$v \text{ (m s}^{-1}\text{)}$	$\frac{v}{c}$	γ
3.00×10^2	0000001	1.000000000
3.00×10^5	000100	1.0000005
3.00×10^7	0.100	1.005
1.50×10^8	0.500	1.155
2.60×10^8	0.866	2.00
2.70×10^8	0.900	2.29
2.97×10^8	0.990	7.09
2.997×10^8	0.999	22.4

Example 2 - Long-lived Muons

Standard example

In the Earth's atmosphere, high-energy cosmic rays interact with the nuclei of oxygen atoms 15 km above the surface of the Earth to create a cascade of high-velocity sub-atomic particles.

One of these particles is a **muon**, which is unstable.

The mean lifetime of a stationary muon, as measured by an atomic clock, is 0.000 002 196 s (2.196 μ s). The muons created by cosmic radiation typically travel at 99.97% of the speed of light, so at this speed, Newtonian physics would predict that a muon would travel about 659 m.

$$\begin{aligned}
 s &= v\Delta t \\
 &= (0.9997)(3 \times 10^8)(2.196 \times 10^{-6}) \\
 &= 659 \text{ m}
 \end{aligned}$$

Given that, after 10 lifetimes, you can expect there to be essentially no muons remaining, then after a distance of 6.58 km, or *at a height of about 8.42 km above the surface of Earth, you would expect that no muons would be detected.*

However, you find that muons created by cosmic radiation are actually detected at the surface of the Earth. This means that the fast-moving muons have existed for a much longer period of time than they should have. A muon that strikes the surface would have existed at least 22.8 times its predicted lifespan as a stationary muon.

From the table above, the muon would be travelling at least 0.999c - hence it exists for longer than its normal mean lifetime.

Time dilation provides the explanation to this unusual observation.

The 'normal' mean lifetime of a muon is about $2.2\mu\text{s}$. However, this is the mean lifetime when measured in a stationary frame of reference. Muons travel very fast; in fact a speed as great as $0.999c$ is possible. At this speed, an observer on Earth will see the lifetime of a muon as far greater than $2.2\mu\text{s}$:

$$\begin{aligned}
 t &= t_0 \gamma \\
 &= \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}} \\
 &= \frac{2.2 \times 10^{-6}}{\sqrt{1 - \frac{(0.999c)^2}{c^2}}} \\
 &= \frac{2.2 \times 10^{-6}}{\sqrt{1 - 0.999^2}} \\
 &= 49.21 \mu\text{s} \text{ (which is more than 22 times as long as in the stationary frame)}
 \end{aligned}$$

Handwritten notes and calculations:

Observer (lab) $t = ?$

$0.9997c$

$t_0 = 2.196 \mu\text{s}$

$$t = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{2.196 \times 10^{-6}}{\sqrt{1 - \frac{(0.9997c)^2}{c^2}}} = 8.97 \times 10^{-5} \text{ s}$$

An observer on Earth would see the muon's time run much slower. The slower time means that many muons live long enough to reach the Earth's surface.

EXTENSION

The Twin Paradox

If Clare sees time for Amaya and Binh running slowly, then Amaya and Binh will age slowly. But if Amaya and Binh see that time for Clare has slowed down, then Clare will age more slowly. So what happens when Amaya and Binh decide to turn their spaceship around and come home? Who will have aged more?

To solve this paradox, or contradiction, Einstein described a thought experiment in which one of a set of twins heads off on a long space journey, while the other twin stays on Earth.

The travelling twin finds that when she returns, her remaining twin has become quite elderly (refer to Figure 8.2.6). Although each twin is in constant motion relative to the other, they both see the other twin ageing more slowly. So why did the twin on the spaceship age more slowly than the twin on Earth?

The key to this paradox is that only one twin has spent the entire time in a single inertial (non-accelerating) frame of reference. The other twin spent some time in

two different inertial frames of reference, one on the way out and the other on the way back. The fact that the travelling twin experienced at least three different periods of acceleration or deceleration has very little effect on the difference in time for the two twins. This means that Einstein's theory of general relativity is not the major factor in the explanation of the twin paradox.

The explanation can be found by considering the Doppler effect on the frequency at which the clock signals sent by each twin are received by the other twin. Consider the situation of twins, Linda, who likes to stay at home, and Yvette, who likes to travel. Yvette leaves on a trip in a spaceship travelling at 80% of the speed of light. Each twin has a clock; but, instead of having standard clocks, each twin has a clock that sends out a light signal once a year. Linda's clock sends out a pulse of light at midnight on New Year's Eve that would eventually catch up to Yvette as she travels away from Earth. Yvette's clock also sends out a light signal at the start of her year.

There is a distinction that must be made here between the time that each twin experiences—Yvette's time will be dilated time and Linda's time will be proper time. These distinctions are not normally made, because you can't tell which one of two inertial frames of reference is moving relative to the other. In this case, however, we know that Yvette is the one travelling and therefore is observed to live in dilated time, while Linda is not moving and lives in proper time. We know this because in Yvette's spaceship there is a chandelier that was observed to hang down at an angle from the vertical during certain stages of the trip. This indicates that Yvette's spaceship was accelerating or decelerating, so she was the one moving. In a spaceship travelling at 80% of the speed of light, it would take three of Yvette's dilated years for the first of stay-at-home Linda's time signals to reach her travelling sister. Similarly, it would take three of Linda's proper years for Yvette's first time signal to reach the twin on Earth. This is consistent

with the idea of relative motion, and that neither inertial frame of reference is special.

In the time that the travelling twin Yvette travels for 15 of her dilated years, she would have received five yearly time signals from the stay-at-home twin Linda. Linda would have received five time signals from Yvette in 45 proper years. At this point, Yvette decides to come home. Yvette turns her spaceship around and she is now in a different inertial frame of reference, but it will still take 15 of her dilated years to get home. Now, as Yvette is travelling back towards the approaching time signals sent by Linda, she will receive three time signals from Linda during each one of her dilated years. So in the 15 dilated years it takes for Yvette to travel back, she will receive 45 yearly signals from Linda. Linda, on the other hand, will receive three time signals from Yvette each proper year, so she will receive a total of 15 yearly signals in five proper years. The total trip for Yvette took 30 dilated years, 15 out and 15 back, so Yvette has aged by 30 years. Linda remained on Earth and had 50 birthdays, so she is now 20 years older than Yvette.

Although it is often called a paradox, there is actually nothing impossible or illogical about this story. Einstein himself pointed out that, due to the Earth's rotation, and therefore centripetal acceleration, a clock on the Earth's equator would run a little more slowly than one at the poles. This has now actually been found to be the case. In fact, in 1971 accurate atomic clocks were flown around the world on commercial flights. When compared with those left behind, the difference of about a quarter of a microsecond was just what Einstein's theory predicted. Now there are many satellites in orbit around the Earth, so the theory has been well and truly tested many times. Indeed, global positioning systems must take the relativistic corrections into account to ensure their accuracy.

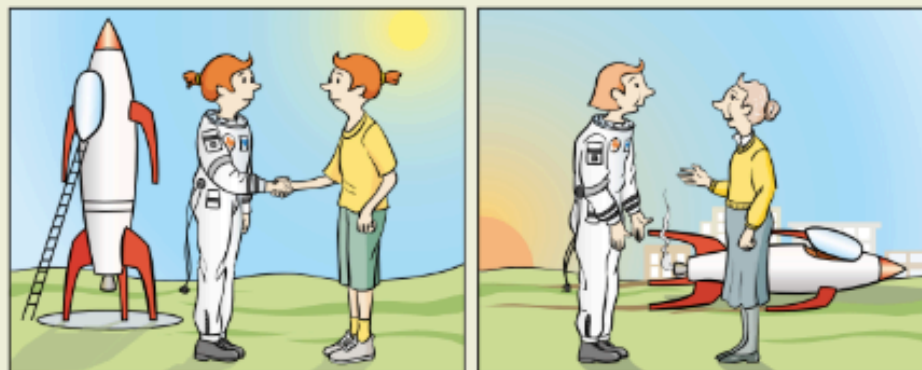


FIGURE 8.2.6 The Twin Paradox describes the phenomenon where one twin ages less quickly than the other after travelling in a non-inertial frame.



 Observer (twin on Earth)

$$t = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$= 25 \text{ yrs} \quad \therefore \text{She is older.}$$



 $t_0 = 1.0 \text{ yr}$

 $v = 0.9999c$

Length Contraction

In the previous section, a photon travelling from the top of the space ship to the floor was used to compare the period of time for an event. It was deliberately set up in the spaceship so that this light path, of distance d , was **perpendicular** to the velocity. The reason for this was that we could expect distances in this perpendicular direction to be unaffected, but this may not be the case for distances in the direction of travel.

Einstein showed that while **perpendicular distances are unaffected**, *relative motion affects lengths in the direction of travel*.

Einstein showed that as a person observes a fast-moving object, its length or the distance it travels in a certain time, will appear shorter than the distance seen when the object is stationary in the observer's reference frame.

It will be *shorter by the same factor by which time appears to be extended* and so Clare would see the length of the space ship contracted when compared to the length as measured by Angela and Bill.

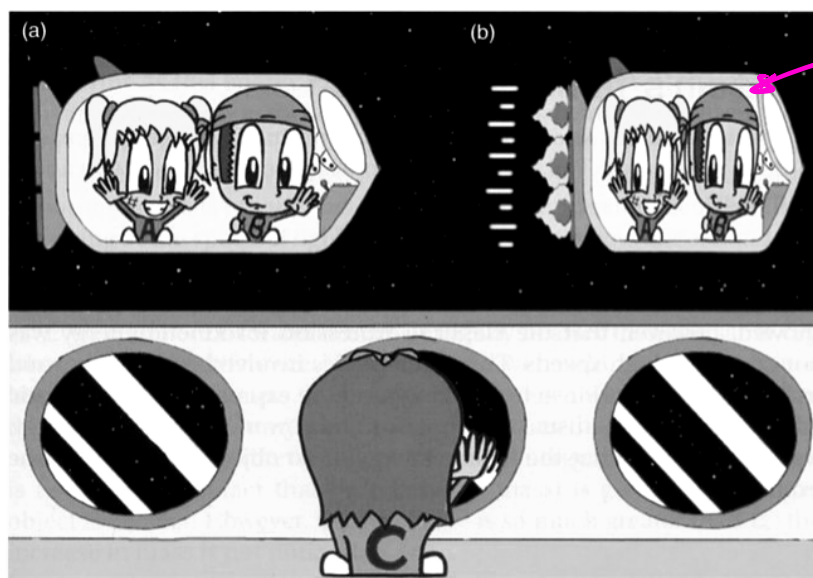
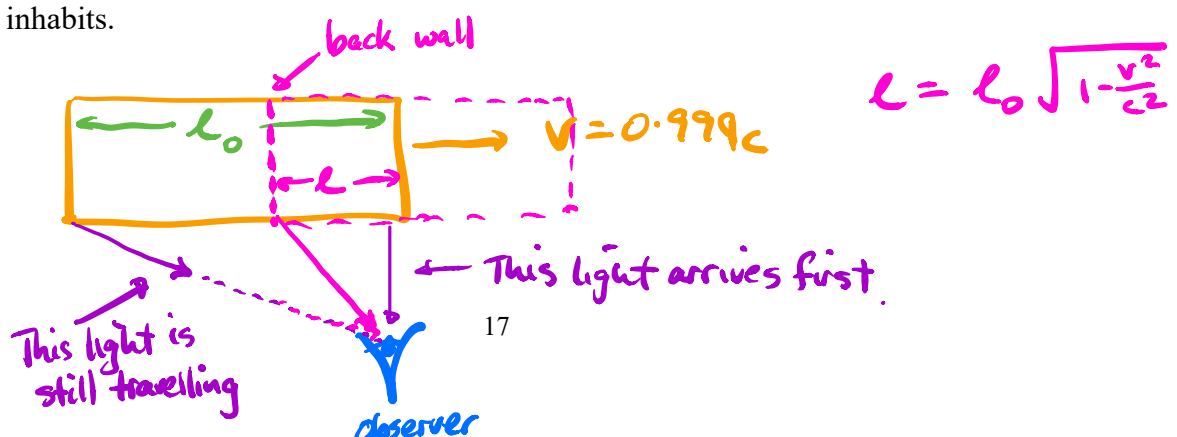


Figure 6.32

(a) Clare watches as Angela and Bill prepare for blast off. (b) As Angela and Bill speed past, Clare sees their rocket ship foreshortened, but the other dimensions remain the same.

Remember that **length contraction occurs only in the direction of travel**, not in any perpendicular direction. To Clare, Angela and Bill's spaceship will appear foreshortened, but its width and height will remain unaltered.

These two simple notions - length contraction and time dilation - are at the core of the special theory of relativity. Together they tell Physicists about spacetime - the four-dimensional world that humankind inhabits.



Einstein's length contraction equation shows that an object with a proper length of l_0 , when measured at rest, will have a shorter length l , parallel to the motion of its moving frame of reference when measured by an observer that is in a stationary frame of reference.

Length contraction can be represented as:

$$l = \frac{l_0}{\gamma}$$

This becomes:

$$l = l_0 \sqrt{1 - \frac{v^2}{c^2}}$$

On data sheet.

An example of length contraction is shown with a tennis ball in Figure 8.3.6. The length in the direction of the motion is contracted, but the height is not.

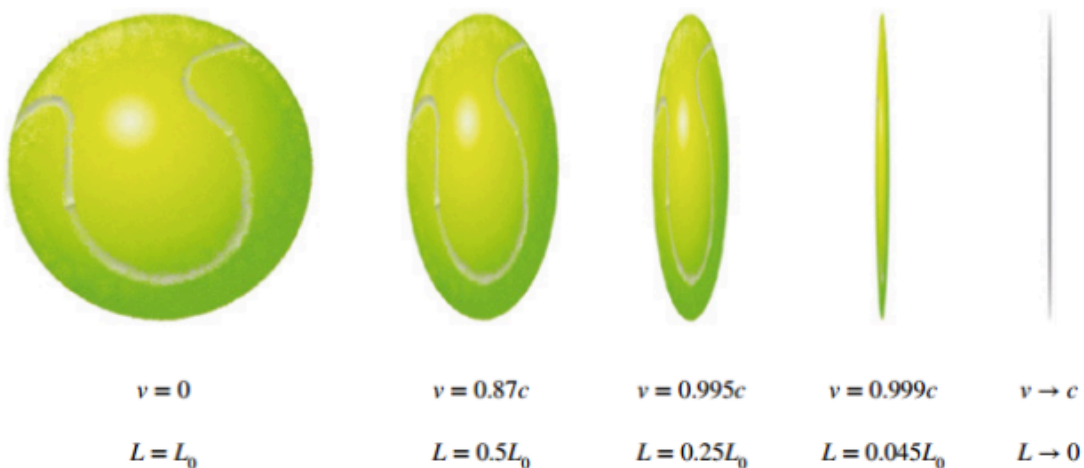


FIGURE 8.3.6 As the tennis ball moves faster to the right, its length in this dimension is contracted, but its height and depth remain the same.

Example 1

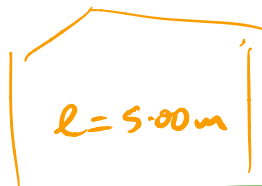
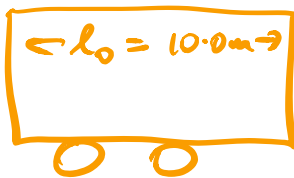
▲ Assume <i>Gedanken</i> conditions exist in this question. A stationary observer on Earth sees a very fast car travelling by at $2.50 \times 10^8 \text{ m s}^{-1}$. When stationary, the car is 3.00 m long. Calculate the length of the car as seen by the stationary observer. Use $c = 3.00 \times 10^8 \text{ m s}^{-1}$. 3 parts	
Thinking Identify the variables: the length for the stationary observer is l, the proper length of the moving car is l_0 and the velocities are v and the constant c.	Working $l = ?$ $l_0 = 3.00 \text{ m}$ $v = 2.50 \times 10^8 \text{ m s}^{-1}$ $c = 3.00 \times 10^8 \text{ m s}^{-1}$
Thinking Use Einstein's length contraction formula and the Lorentz factor.	Working $l = \frac{l_0}{\gamma}$ $l = l_0 \sqrt{1 - \frac{v^2}{c^2}}$
Thinking Substitute the values for l_0, v and c into the equation and calculate the answer l.	Working $l = (3.00) \sqrt{1 - \frac{(2.50 \times 10^8)^2}{(3.00 \times 10^8)^2}}$ $= (3.00)(0.553)$ $= 1.66 \text{ m}$

$0.833c$

l_0 in the moving frame

$$\begin{aligned}
 l &= l_0 \sqrt{1 - \frac{v^2}{c^2}} \\
 &= 3.00 \sqrt{1 - \frac{(0.833c)^2}{c^2}} \\
 &= \underline{1.66 \text{ m}}
 \end{aligned}$$

Question I want a 10.0 m car to fit a 5.0 m garage.



$v = ?$

Observer

$$l = l_0 \sqrt{1 - \frac{v^2}{c^2}}$$

$$\Rightarrow \frac{5.00}{10.0} = \sqrt{1 - \frac{v^2}{c^2}}$$

$$\Rightarrow 0.25 = 1 - \frac{v^2}{c^2}$$

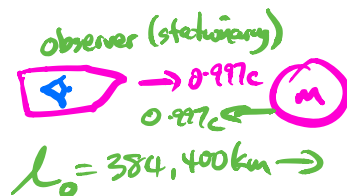
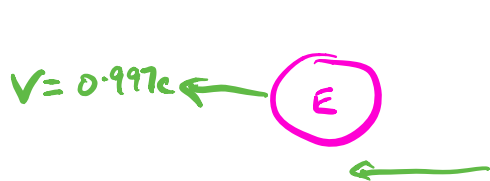
$$\Rightarrow \frac{v^2}{c^2} = 0.75$$

$$\begin{aligned}
 \Rightarrow v &= \sqrt{0.75 c^2} \\
 &= \underline{0.866 c}
 \end{aligned}$$

Example 2

LENGTH CONTRACTION FOR DISTANCE TRAVELLED

Assume <i>Gedanken</i> conditions exist in this example. A spaceship is travelling at $0.997c$ from Earth to the Moon. The proper distance from the Earth to the Moon is $384\,400\text{ km}$. When the pilot looks out of the window, the distance between the Earth and the Moon looks much less than that. Calculate the distance that the pilot sees.	
Thinking	Working
Identify the variables: the length seen by the pilot is L , the proper length of the distance is L_0 and the velocity is v .	$L = ?$ $L_0 = 384\,400\text{ km}$ $v = 0.997c\text{ ms}^{-1}$
Use Einstein's length contraction formula and the Lorentz factor.	$L = \frac{L_0}{\gamma}$ $= L_0 \sqrt{1 - \frac{v^2}{c^2}}$
Substitute the values for L_0 and v into the equation. Cancel c and calculate the answer, L .	$L = 384\,400 \times \sqrt{1 - \frac{(0.997c)^2}{c^2}}$ $= 384\,400 \times \sqrt{1 - (0.997)^2}$ $= 384\,400 \times 0.0774$ $= 29\,753$ $= 29\,800\text{ km}$



Observer is on the ship
 $\Rightarrow$ Ship is stationary
 $\therefore$ E + M are moving
 $\Rightarrow L_0 = 384\,400\text{ km}$

$$\begin{aligned}
 l &= l_0 \sqrt{1 - \frac{v^2}{c^2}} \\
 &= (384\,400) \sqrt{1 - \frac{(0.997c)^2}{c^2}} \\
 &= 29\,800\text{ km}
 \end{aligned}$$

Relativistic Mass and Momentum

Calculations involving time and distance get a bit rubbery when one considers relativistic effects like time dilation and length contraction. It would not be surprising, therefore, to hear that

momentum also displays relativistic effects.

$\Rightarrow$ mass shows relativistic effects

If a rocket ship is travelling at 99% of c , why can't it simply turn on its rocket motor and accelerate up to 100%, or more, of c ?

A full answer to this question was not given in Einstein's original 1905 paper on relativity. Some years later, he showed that as the speed of a rocket ship approaches c , its momentum increases as one might expect, but this is not reflected in a corresponding increase in speed.

While the force of the rocket engine increases the momentum, near the speed of light the increase in momentum results in **smaller and smaller increases in speed**. So no amount of impulse can accelerate the rocket ship to the speed of light.

One interpretation of this strange behaviour of momentum is that **it appears that the inertial mass of the rocket ship is increasing**.

Although Einstein preferred to stay with relativistic momentum, **it is helpful to think in terms of the mass of the rocket ship increasing as its speed approaches c .**

Hence, in terms of Newton's law, one can simply say that **as the speed increases, the force required to produce a given acceleration increases towards infinity, because the mass approaches infinity**.

Thus, no matter how much force is applied or work done, **a rocket ship (or anything else) can never reach the speed of light**. In fact **only things without mass** can travel at the speed of light, like **photons**.

The relativistic momentum is given by:

$$p = \gamma p_0$$

$p = m v$ where $v = \text{relativistic}$
 $p_0 = m_0 v$ where $v = \text{normal}$.

This becomes:

$$p = \frac{p_0}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{m_0 v}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$p = \frac{p_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

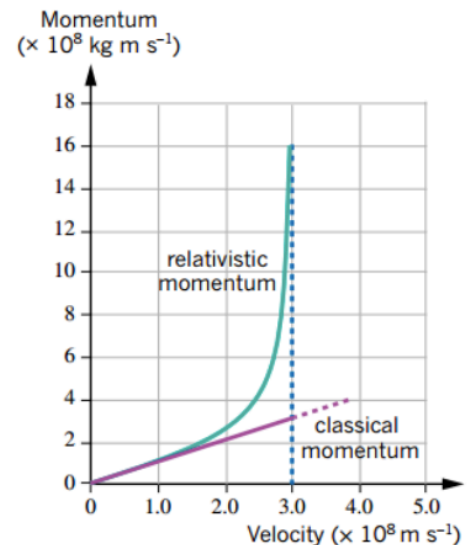


FIGURE 8.4.2 The relationship between classical momentum and velocity, and the relationship between relativistic momentum and velocity, for a 1 kg mass.

$u' = 0.8c$



$v = 0.9c$



⑤

Example

Assume *Gedanken* conditions exist in the following questions.

- ▲ (a) Calculate the momentum, as seen by a stationary observer, provided to a rocket ship with a rest mass of 1000 kg, as it goes from rest up to a speed of $0.990c$. 3 parts

Thinking

Identify the variables: the rest mass is m_0 , and the velocity of the rocket ship is v .

Working

$$\begin{aligned}\Delta p &=? \\ m_0 &= 1000 \text{ kg} \\ v &= 0.990c\end{aligned}$$

Thinking

Use the relativistic momentum formula.

Working

$$p = \gamma m_0 v$$

Thinking

Substitute the values for m_0 and v into the equation and calculate the answer p .

Working

$$\begin{aligned}p &= \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} m_0 v \\ &= \frac{1}{\sqrt{1 - \frac{(0.990c)^2}{c^2}}} (1000)(0.990 \times 3.00 \times 10^8) \\ &= \frac{1}{\sqrt{1 - 0.990^2}} (1000)(2.97 \times 10^8) \\ &= (7.08881)(1000)(2.97 \times 10^8) \\ &= 2.11 \times 10^{12} \text{ kg m s}^{-1}\end{aligned}$$

- ▲ (b) If twice the relativistic momentum from part (a) is applied to the stationary rocket ship, calculate the new final speed of the rocket ship in terms of c . 3 parts

Thinking

Identify the variables: the rest mass is m_0 , and the relativistic momentum of the rocket ship is p .

Working

$$\begin{aligned}p &= 2 \times (2.11 \times 10^{12}) \\ &= 4.21 \times 10^{12} \text{ kg m s}^{-1} \\ m_0 &= 1000 \text{ kg} \\ v &=?\end{aligned}$$

Thinking

Use the relativistic momentum formula, rearranged.

Working

$$\begin{aligned}p &= \gamma m_0 v \\ &= \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} m_0 v \\ v &= \frac{p}{m_0 \sqrt{1 + \frac{p^2}{m_0^2 c^2}}}\end{aligned}$$

Thinking

Substitute the values for m_0 and p into the rearranged equation and calculate the answer v .

Working

$$\begin{aligned}v &= \frac{p}{m_0 \sqrt{1 + \frac{p^2}{m_0^2 c^2}}} \\ &= \frac{(4.21 \times 10^{12})}{(1000) \sqrt{1 + \frac{(4.21 \times 10^{12})^2}{(1000)^2 (3.00 \times 10^8)^2}}} \\ &= \frac{(4.21 \times 10^{12})}{(1000)(14.07)} \\ &= 2.99 \times 10^8 \text{ m s}^{-1} \\ &= 0.997c\end{aligned}$$

Relativistic Addition of Velocities

No two objects can have a relative velocity greater than c . Remember - the speed of light is the same in all inertial (non-accelerating) reference frames.

Consider the following situation.

A spacecraft moving at $0.9c$ fires a missile at $0.8c$ forwards and relative to it. What does a stationary observer measure as the speed of the missile?

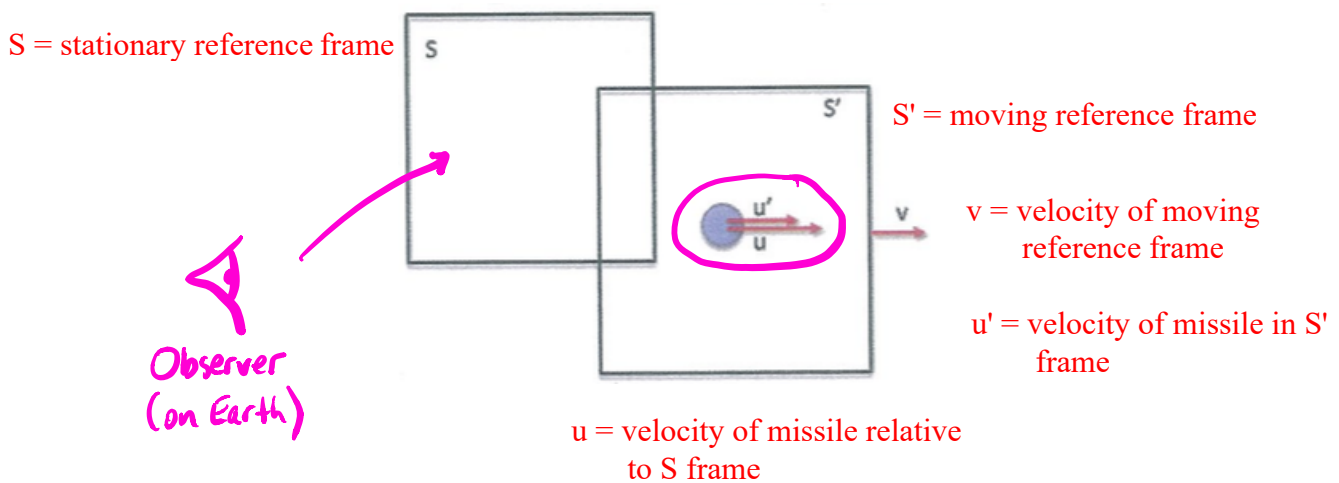
According to classical Physics, a stationary observer sees this as $1.7c$, which is impossible.

Velocities must transform according to the Lorentz transformation, leading to a non-intuitive result called ***Einstein velocity addition***.

(See Ch. 8.4 - Relativistic addition of velocities.)

Solution to the "problem"

Consider two observers in relative motion to each other and both are observing the motion of the missile.



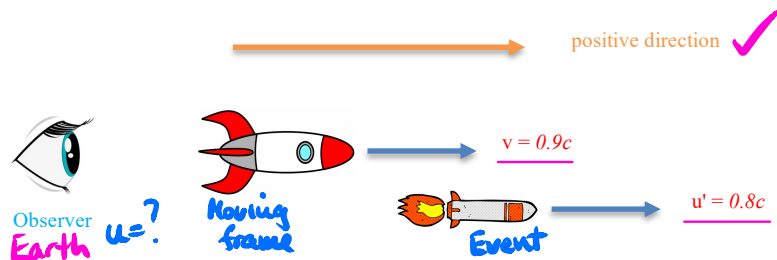
Note: Consider the motion of the missile as an "event" that is moving relative to the stationary frame. In this case, the speed of the event is u .

The equation describing u is given by:

$$u = \frac{u' + v}{1 + \frac{u'v}{c^2}}$$

where

- u = velocity of the moving object in the stationary frame
- u' = velocity of the moving object in the moving frame
- v = velocity of the moving reference frame
- c = speed of light



In this situation: $u' = 0.8c$, $v = 0.9c$, $u = ?$

i.e.

$$u = \frac{u' + v}{1 + \frac{u'v}{c^2}}$$

$$= \frac{0.8c + 0.9c}{1 + \frac{(0.8c)(0.9c)}{c^2}} \quad \left. \vphantom{\frac{0.8c + 0.9c}{1 + \frac{(0.8c)(0.9c)}{c^2}}}\right\} \text{Consistent}$$

$$= \frac{1.7c}{1.72} \quad \leftarrow (< 1.00c)$$

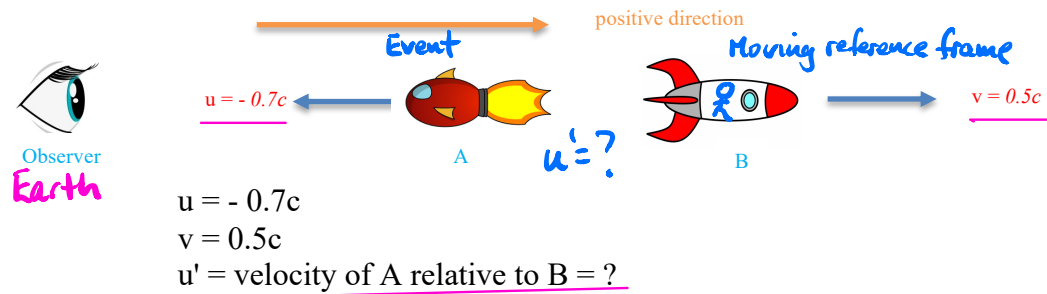
$$= \underline{0.988c} \quad \checkmark$$

Example 1

Two spaceships are moving apart from each other, A at $0.7c$ and B $0.5c$ in the opposite direction. What is the velocity of A as seen by an observer in B?

Consider B as the moving reference frame relative to Earth (assumed stationary).
Consider A as an "event".

Take the direction of B as +ve.



$$u' = \frac{u - v}{1 - \frac{uv}{c^2}}$$

$$= \frac{-0.7c - 0.5c}{1 - \frac{(-0.7c)(0.5c)}{c^2}} \quad \left. \vphantom{\frac{-0.7c - 0.5c}{1 - \frac{(-0.7c)(0.5c)}{c^2}}}\right\} \text{Be consistent}$$

$$= \frac{-1.2c}{1.35} \quad \leftarrow$$

$$= \underline{-0.889c} \quad \checkmark$$

$\therefore$ A is receding at $0.889c$ from B

Note If the direction of A is taken as +ve, $u' = 0.889c$. The +ve sign still means that A is receding from B.

In the following diagram, Amaya throws a ball in the moving reference frame to Binh at a speed u' - this is the speed measured in their reference frame.

Clare is stationary and observes their reference frame moving at speed v . She measures the speed of the ball as u .

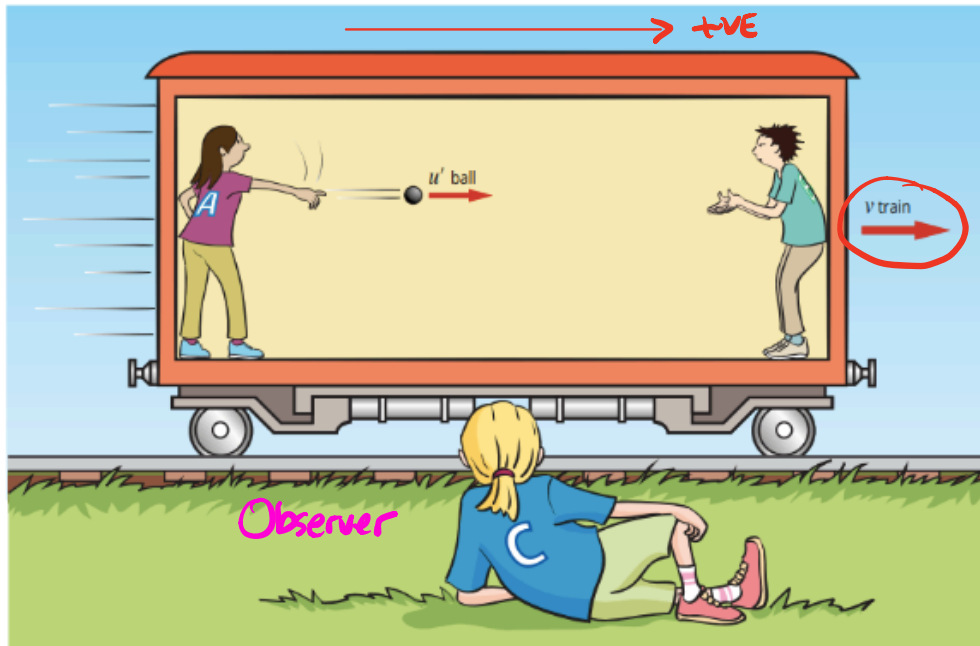


FIGURE 8.4.3 The velocity of the ball relative to the train carriage is u' , the velocity of the ball for the stationary observer (Clare) is u , and the velocity of the train carriage is v .

Example 2

RELATIVISTIC ADDITION OF VELOCITIES

Assume *Gedanken* conditions exist in this example. An observer on a train platform sees a train carriage passing by travelling at $0.90c$. Inside the carriage a ball is thrown towards the front of the carriage at $0.20c$. Calculate the velocity of the ball as seen by the stationary observer, in terms of c .

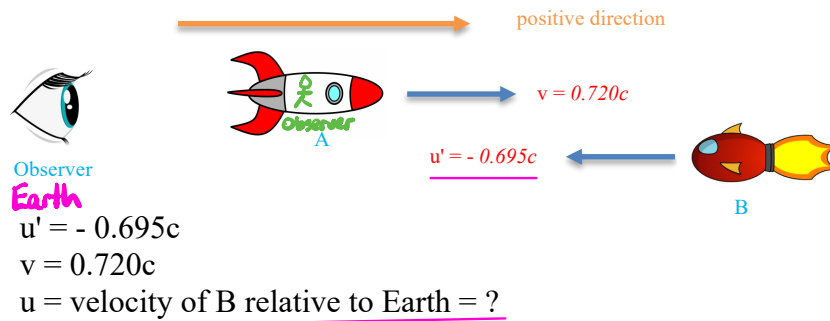
Thinking	Working
Identify the variables for the velocity for the stationary observer u , the velocity of the ball relative to the carriage u' and the velocity of the carriage v .	$u = ?$ $u' = 0.20c$ $v = 0.90c$
Use the relativistic velocity addition formula.	$u = \frac{u' + v}{1 + \frac{u'v}{c^2}}$
Substitute the values for u' and v into the equation and calculate the answer u .	$ \begin{aligned} u &= \frac{(0.20c) + (0.90c)}{1 + \frac{(0.20c)(0.90c)}{c^2}} \\ &= \frac{(1.10c)}{1 + \frac{(0.18c^2)}{c^2}} \\ &= \frac{1.10c}{1 + 0.18} \\ &= \frac{1.10c}{1.18} \\ u &= 0.932c \end{aligned} $

Example 3 (Q16 - 2018 WACE Examination)

In the future it is possible that humans may travel to distant places like Alpha Centauri that is 4.13×10^{13} km from Earth. Imagine you are on a spacecraft travelling past Earth towards Alpha Centauri at $0.720c$ relative to Earth. On your journey you pass another spacecraft travelling parallel and in the opposite direction to you. You measure the relative velocity of the other spacecraft as $0.695c$.

- (a) Calculate the velocity of the other spacecraft relative to Earth.

(3 marks)



$$\begin{aligned}
 u &= \frac{u' + v}{1 + \frac{u'v}{c^2}} \\
 &= \frac{-0.695c + 0.720c}{1 - \frac{(-0.695c)(0.720c)}{c^2}} \\
 &= \frac{0.0250c}{0.500} \\
 &= 0.0500c \quad \checkmark
 \end{aligned}$$

$\therefore$ B is moving at $0.0500c$ relative to Earth

- (b) Calculate the number of years for your spacecraft to journey from Earth to Alpha Centauri as measured by an observer on Earth.

(3 marks)

$$\begin{aligned}
 v &= \frac{s}{t} \\
 \Rightarrow t &= \frac{s}{v} \\
 &= \frac{4.13 \times 10^{16}}{(0.720)(3.00 \times 10^8)} \quad \leftarrow \text{conversion} \\
 &= 1.912 \times 10^8 \text{ s} \\
 &= \underline{6.06 \text{ years}} \quad (\text{taking } 1.00 \text{ year} = \underline{365.25 \text{ days}})
 \end{aligned}$$

- (c) Calculate the number of years the journey will take as measured by those on the spacecraft travelling at $0.720c$ relative to Earth. (3 marks)

$$t = 6.06 \text{ years}$$

$$t_0 = ?$$

→ moving reference frame
⇒ Find t_0

$$t = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\Rightarrow t_0 = t \sqrt{1 - \frac{v^2}{c^2}} \quad \checkmark$$

$$= (6.06) \sqrt{1 - \frac{(0.720)^2}{c^2}}$$

$$= \underline{4.20 \text{ years}} \quad \checkmark$$

- (d) For those on the spacecraft travelling to Alpha Centauri at $0.720c$ relative to Earth, calculate the time they would have observed to have elapsed on Earth during the journey from Earth to Alpha Centauri. (3 marks)

$$t = 4.20 \text{ years}$$

$$t_0 = ?$$

The Earth is the moving reference frame.
⇒ Find t_0

$$t = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\Rightarrow t_0 = t \sqrt{1 - \frac{v^2}{c^2}} \quad \checkmark$$

$$= (4.20) \sqrt{1 - \frac{(0.720)^2}{c^2}}$$

$$= \underline{2.91 \text{ years}} \quad \checkmark$$

Einstein's Equation

As the momentum of an object increases, so does the energy. The classical relationship between the two can be written as:

$$E_k = \frac{1}{2}mv^2 = \frac{1}{2}pv$$

As v approaches relativistic speeds near c , the kinetic energy, as well as the momentum, will increase towards infinity.

Einstein showed that the classical expression for kinetic energy was not correct at high speeds.

The mathematics involved is beyond this course but Einstein, working from the expression for relativistic momentum and the usual assumptions about work, forces and energy, was able to show that the kinetic energy of an object was given by the expression:

$$E_k = (\gamma - 1)m_0c^2$$

where E_k = the kinetic energy of an object corrected for relativistic effects (J).
 γ = the Lorentz factor, which is the factor that converts rest mass to relativistic mass.
 m_0 = the mass of an object when it is at rest in our frame of reference (kg).
 c = the speed of light ($3.00 \times 10^8 \text{ ms}^{-1}$).

Einstein's expression can be rewritten as:

$$\begin{aligned} E_k &= (\gamma - 1)m_0c^2 \\ &= \gamma m_0c^2 - m_0c^2 \\ \Rightarrow \gamma m_0c^2 &= E_k + m_0c^2 \end{aligned}$$

Einstein interpreted this expression as being an expression for the **total energy**, of the object.

On the data sheet,

$$E_T = \frac{mc^2}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$E_{\text{total}} = E_{\text{kinetic}} + E_{\text{rest}}$$

On the data sheet:

$$E = mc^2$$

$$\Rightarrow E_T = \gamma m_0c^2 = E_k + m_0c^2$$

Now as γm_0 is equal to the **relativistic mass (m)**, the total energy can simply be written as:

$$E_T = mc^2 = E_k + m_0c^2$$

where $E_k = \frac{1}{2}mv^2$

This is the relativistic E_k

NOTE The "m" term is NOT m_0 or γm !

Einstein wondered - what is the significance of the rest energy, the m_0c^2 term?

As this energy depends directly on the mass of the object, it appears to be energy associated with mass, and so mass, he realised, was a form of energy.

Physicists now understand that this equation shows that mass and energy are totally interrelated.

Example 1

A cup of hot water is ever so slightly heavier than the same cup when cold. The difference is far less than one could possibly measure, but it is.

The heat means the molecules have more kinetic energy, and as the total energy is the sum of the rest and kinetic energy, the total energy (mc^2) is greater.

This is reflected in the fact that m (relativistic mass) is greater, so a hotter object is heavier.

Since m_0c^2 is so much greater than E_k , the increase in mass is not noticeable.

Example 2

▲ Calculate how much more massive a 200 g cup of boiling water is than the same cup of iced water, given that the specific heat capacity of water is $4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$. 5 parts	
Thinking Identify the variables for calculating the increase in energy (in this case, the increase in the internal energy of the water).	Working $m = 0.200 \text{ kg}$ $c_w = 4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$ $\Delta T = 100 \text{ K}$
Thinking Use the appropriate formula (in this case, the specific heat formula).	Working $Q = mc_w \Delta T$
Thinking Substitute the values into the equation and calculate the energy (in this case, Q).	Working $Q = (0.200)(4.18 \times 10^3)(100)$ $= 8.36 \times 10^4 \text{ J}$
Thinking Use the total energy formula.	Working $E = mc^2$
Thinking Equate the energy values (if necessary), substitute the values for E and c into the equation, and calculate the answer m.	Working $m = \frac{E}{c^2} = \frac{Q}{c^2}$ $= \frac{(8.36 \times 10^4)}{(3.00 \times 10^8)^2}$ $= 9.29 \times 10^{-13} \text{ kg}$ (This increase in mass is very tiny.)

EXAMPLE 2

Calculate the relativistic kinetic energy of a proton travelling at $0.750c$.

$$E_T = E_K + E_0$$

$$\Rightarrow E_K = E_T - E_0$$

$$= \frac{m_0 c^2}{\sqrt{1 - \frac{v^2}{c^2}}} - m_0 c^2$$

$$= \left[\frac{1}{\sqrt{1 - \frac{(0.750c)^2}{c^2}}} - 1 \right] (1.67 \times 10^{-27}) (3.00 \times 10^8)^2$$

$$= \underline{7.69 \times 10^{-11} \text{ J}}$$

NOTE: This can be written as -

$$E_K = (\gamma - 1)m_0 c^2$$

PHYSICS IN ACTION

Particles gaining mass

When an object moves in a circular path, it does so as a result of a centripetal force that acts towards the centre of the circular path. Centripetal force therefore acts continuously at a right angle to the velocity of the object. There are a number of actions that could cause the centripetal force on an object, such as the tension in a string tied to a rubber stopper or the gravitational force of the Earth on the Moon.

Another action that causes circular motion is the force on a charged particle that is moving at right angles to a magnetic field. The equation that represents the relationship between the magnetic force (F_B) and the centripetal force (F_C) is:

$$\begin{aligned}F_B &= F_C \\qvB &= \frac{mv^2}{r} \\r &= \frac{mv}{qB} \\r &= \frac{m}{qB} \times v\end{aligned}$$

The final relationship shows that, if the mass m , charge q and magnetic field B are all constant, then the radius of the circular path is directly proportional to the velocity of the charged particle. So, theoretically, if the velocity increases by a factor of 2, then the radius will also increase by a factor of 2. However, this is not the case.

In circular accelerating devices, such as cyclotrons and the Australian Synchrotron, it is evident that, as the velocity of a charged particle increases, the radius of its path also increases, but to a much greater degree than that expected. According to the relationship shown above, if the charge q and the magnetic field B don't change, then the only explanation for the extra increase in radius is that the mass of the particle m must increase.

In fact, the mass of an electron travelling at 99.99999% of the speed of light seems to increase to 6000 times the mass of an electron at rest. There is no explanation for this phenomenon in Galileo's relativity or Newtonian physics. This will all be explained later in the chapter, using Einstein's theories.

EXTENSION

Gravity waves

Einstein's general theory of relativity predicted the existence of gravity waves that carry energy and angular momentum as ripples through spacetime at the velocity of light. These waves were considered capable of deflecting a beam of light (photons) from a distant star when it was in the gravitational field of a large mass such as the Sun. The deflection calculated using Einstein's theory gave an answer that was greater than that from Newton's gravity; a result not unexpected because Newton's physics was developed around non-relativistic speeds.

Einstein's theory was purely theoretical and at that time had no empirical evidence to support it. Results obtained in 1919 from the observation of a solar eclipse gave empirical evidence that light from a distant star underwent deflection as predicted by Einstein's calculations. These results were seriously questioned at the time, but were substantiated by Professor Alexander Ross, professor of physics at the University of Western Australia, when he led a group to do similar observation of a solar eclipse in 1922 at Wallal near Broome, Western Australia.

In 1974, observations of a pair of neutron stars in close orbit about each other were made by Russell Hulse and Joe Taylor from the University of Massachusetts. They found that the period of orbit of this star pair (measured extremely precisely) about each other was decreasing.

These observations agreed with the predictions of Einstein's theory that orbiting stars would spiral in towards each other as they radiated gravity waves. Taylor and Hulse received the Noble Prize in Physics in 1993 for the work they did on the energy loss from binary pulsars and their resultant proof from this of the existence of gravity waves.

In 1995 Australia established a national consortium involving the Australian National University, the University of Western Australia, Adelaide University and the CSIRO to develop laser interferometry technology and join a world consortium, called LIGO (Light Interferometer Gravitational Observatory), searching for gravity waves. The Australian facility was set up at Gingin, 100 km north of Perth, in 1999. In September 2015 success was achieved by LIGO when gravity waves were detected almost simultaneously at the USA sites of Hanford and Livingston. Fifty-six Australian scientists were involved in this discovery.

EXTENSION

Real-world relativity

The global positioning system

The global positioning system (GPS) is a satellite system that is now part of everyday life. For it to function accurately, adjustments need to be made to its component satellites to compensate for the effects of both special and general relativity. There are many uses for GPS including aircraft and motor vehicle navigation, driverless vehicles, crop planting, spraying and fertilising, both military and civilian drones, and the control of autonomous trucks in the mining industry. The GPS device provides accurate real-time information on position, speed and direction of travel on the Earth's surface.

The GPS system is configured using a network of satellites that are in various orbits around the Earth at an altitude of just over 20 000 km, a speed of around 3900 m s^{-1} and a period of around 12 hours (Figure 8.1.10). The satellites are placed in various orbital planes that are set so that four or more of them are visible to any point on the Earth's surface at any one time. Their orbits are not geosynchronous; that is, they do not stay above the same point on the Earth's surface.

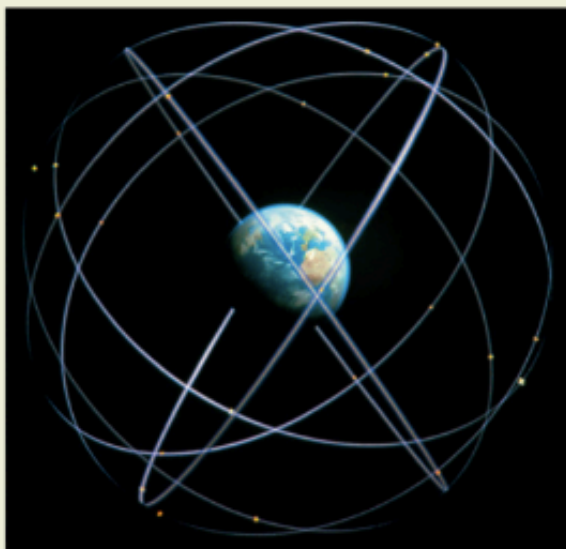


FIGURE 8.1.10 Global positioning satellite orbits.

Each satellite is fitted with an atomic clock that has accuracy in the order of one nanosecond per day. It is the comparison of the time signal received from each satellite visible to the GPS device that determines the position, speed and course of the device. The resultant data received is compared based on the accurately known position of each satellite by a process called trilateration and an absolute position on the Earth's surface determined within 5–10 metres in just a few seconds.

For positional accuracy to be achieved, the time on each GPS satellite must be known to within 20–30 nanoseconds. To achieve such accuracy at the speed and height at which these satellites are travelling, the effects of both special and general relativity must be taken into consideration. The satellite is travelling at a high velocity relative to any point on the Earth's surface, so special relativity would have the GPS device 'see' the satellite's atomic clock running slower than the same clock on Earth (time dilation). Calculations show this to be about $7 \mu\text{s}$ per 24 hours.

General relativity predicts that a clock closer to a massive object will 'tick' slower than one that is further away from that object. The implication of this is that the clock in orbit on the satellite will suffer a time contraction (tick faster) relative to the same clock on Earth. The atomic clock on the satellite will be ahead of the atomic clock on Earth. This results in a time gain on the satellite of about $45 \mu\text{s}$ in 24 hours. This phenomenon exists because the curvature of spacetime results in time running slower in a stronger gravitational field. The further you are from a massive object, such as the Earth, the faster time will pass. A discussion of this concept is beyond this course at this time.

Every GPS satellite is therefore affected by two separate relativistic effects with opposite consequences. One gives a gain in 24 hours of around $45 \mu\text{s}$, the other a loss of around $7 \mu\text{s}$. The resultant gain is therefore around $38 \mu\text{s}$ per 24-hour period. Although this does not seem to be much of a difference, in terms of time measurements made on Earth, it introduces a very significant and serious error when applied to the GPS satellites. As the signal travels at the speed of light, the resultant positional error to an Earth-based GPS device would be about 10 km in this $38 \mu\text{s}$ period.

Before each satellite is launched the engineers and technicians who design and work on global positioning systems include parameters that compensate for the general relativistic effects. They do this by slowing down the rate of the satellite's atomic clock. The special relativity requirements are, however, not fully known until the satellite is in its fixed orbit. Once the satellite is in orbit these parameters are adjusted by an inbuilt computer to achieve the necessary clock speed.

Relativity is therefore no longer an abstract theory but a reality applied to many situations.